



The Perceived Ceiling Scale: Validating an Instrument for When “No Borrowing Limit” Starts to Cap Out

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Abstract. Community toy libraries typically advertise unrestricted borrowing — “no limit,” “take what you need” — while relying on volunteers to privately discourage members who test that promise. We report the development and initial validation of the Perceived Lending Ceiling Scale (PLCS), a six-item self-report instrument measuring a member’s subjective sense of when an advertised unlimited-borrowing policy begins to feel bounded. Items were generated from semi-structured interviews with eighteen members across three toy libraries and administered, alongside tenure and library-identifier items, to 247 members across nine independently-run toy libraries not involved in item generation. Exploratory factor analysis supported a single-factor structure explaining 58.4% of variance, with internal consistency ranging from .78 to .87 across libraries (pooled $\alpha = .83$). PLCS scores correlated moderately with the number of informal reminders volunteers logged for each member over the preceding six months ($r = .52$, $p < .001$), providing initial convergent-validity evidence, while showing no reliable association with membership tenure. We discuss the instrument’s use as a diagnostic for organisations that advertise unbounded access while enforcing it informally, and note that a felt ceiling appears to form well before any written one does.

Introduction

Institutions that offer a resource under an unrestricted-access banner rarely mean it in the plain sense of the words. A subscription gym markets unlimited classes while relying on no-show penalties and waitlist priority to keep attendance manageable, and members routinely pay a flat fee that exceeds what a pay-per-visit arrangement would have cost them [1]. The same structure recurs at a much smaller, more local scale: the community toy library, a volunteer-run lending scheme through which toys circulate among children and their carers, typically under a welcome-pack line to the effect that there is no cap on how many items a member may hold at once. What differs at this scale is that the throttle, where one exists, is rarely a system at all — it is a volunteer’s private decision to have a quiet word with a member whose holdings look, to them, excessive.

No validated instrument currently exists for what a member of such a scheme experiences

as they approach that undocumented threshold, as distinct from what they are told the policy is, or from what a volunteer’s own log records after the fact. The libraries-of-things literature has instead measured participation, catalogue composition, and organisational sustainability at the scheme level [2], [3], [4], [5], leaving a member’s own sense of an approaching ceiling unaddressed by any dedicated measure.

This paper reports the development and initial validation of the Perceived Lending Ceiling Scale (PLCS), a brief self-report instrument intended for toy libraries and, we anticipate, related volunteer-run lending schemes that advertise unrestricted access. Our contributions are threefold, each offered with the qualification the early stage of this work warrants:

- We derive a six-item scale from member interviews and report its factor structure across nine independently-run toy libraries, though a wider deployment would strengthen confi-

dence that the structure holds beyond this sample.

- We report internal consistency by library rather than pooled alone, a step we consider informative even though the libraries were not sampled to be regionally representative.
- We provide initial convergent-validity evidence against volunteers' own informal enforcement logs, while noting that such a log is itself an imperfect record of a private conversation, and our results should be read with that limitation in mind.

Related work

Scale-development methodology treats reliability and factor structure as necessary, not sufficient, evidence of validity: item pools are typically generated qualitatively before quantitative refinement, and internal consistency is reported alongside, not instead of, evidence that a scale relates to an external criterion [6], [7]. Convergent validity specifically asks whether an instrument correlates with an independent measure of a related construct, a standard formalised well before self-report scales were common in organisational research [8]. Where an established instrument is shortened for field use, the item-removal process has been shown to threaten validity unless conducted against explicit criteria rather than by convenience [9]; we return to this in our own ablation.

Toy libraries have drawn intermittent scholarly attention as a genre of informal early-childhood infrastructure since at least the early 1990s [10], and more recent work situates them within a broader “library of things” movement extending circulation logics from books to tools, appliances, and toys [2], [3], [4], [5]. This literature documents participation and sustainability at the scheme level; it says comparatively little about an individual member's subjective experience of a scheme's stated access terms.

Advertised-but-informally-bounded access has a closer analogue in consumer research on flat-rate service contracts, where members routinely misjudge the point at which unlimited access stops paying off [1], and in commons research documenting that shared-resource communities frequently enforce their own limits through unwritten social sanctioning rather than posted rules [11]. Self-report measures of behaviour are known to diverge, sometimes substantially,

from logged records of the same behaviour [12], a caution we weigh directly against our own volunteer-log criterion. Broader nonprofit-governance and volunteer-management literatures describe organisations structured much like the ones we study, without addressing member-facing perception directly [13], [14]; queue- and waitlist-based rationing research offers a formal vocabulary for informal allocation in other settings [15].

Two prior Slop University outputs bear directly on this study. [16] found that a posted access statistic in an adjacent community-lending setting predicted little beyond whether members could read the sign it was printed on. [17] found that a resource's stated category failed to predict how quickly it moved through a comparable informal exchange, with a presentation cue accounting for the effect instead. Both caution against taking a single posted or logged figure as a stand-in for member experience, motivating our decision to validate a member-facing measure directly.

Method

Item generation drew on semi-structured interviews with eighteen members recruited across three toy libraries, each asked to describe, in their own words, how they would know if they were “borrowing too much.” Two coders independently open-coded transcripts for candidate ceiling cues — a volunteer's tone, a hold quietly placed on a popular item, a change in how a return was received — and consolidated overlapping cues into an initial pool of eleven items, each phrased as a first-person statement rated on a five-point agreement scale. A pilot administration ($n = 34$, drawn from one further toy library not otherwise sampled) removed three items for cross-loading or floor effects, leaving the six-item pool reported here. Two representative retained items were “I sense a volunteer would rather I didn't borrow anything else this visit” and “I have started leaving items I want on the shelf so I don't look like I'm taking too many.”

The six-item Perceived Lending Ceiling Scale was administered, alongside tenure and library-identifier items, to 247 members across nine independently-run toy libraries not involved in item generation, recruited through each library's own membership list. Composite scores sum the six items (range 6–30). We assessed factor

structure with exploratory factor analysis (principal-axis extraction, oblique rotation) and internal consistency with Cronbach's alpha, computed both pooled and separately within each library:

$$\alpha = \frac{k}{k-1} \left(1 - \frac{\sum_{i=1}^k \sigma_i^2}{\sigma_{\text{total}}^2} \right)$$

where k is the item count, σ_i^2 each item's variance, and σ_{total}^2 the variance of the summed scale.

For convergent validity, we obtained, with each library's and each member's consent, a count of informal reminders — a volunteer's own retrospective log entry recording a private conversation about borrowing volume — issued to that member in the six months preceding the survey. Libraries kept these logs before this study, for their own rostering purposes; none was created for our benefit, a design choice intended to avoid observer effects on the enforcement behaviour itself. We correlated composite PLCS scores against this count (Pearson's r) and, as an ablation, recomputed reliability with the item closest to referencing a volunteer's manner directly removed, to test whether the scale's coherence depended on that one item.

Results

Exploratory factor analysis on the 247-member sample supported a single-factor solution, with the first factor's eigenvalue (3.51) well clear of the second (0.89) and all six items loading above .52. The single factor accounted for 58.4% of total variance. Internal consistency was acceptable to good: pooled alpha reached .83, and library-level alpha ranged from .78 to .87 across the nine libraries, with none falling below the conventional .70 threshold for an exploratory instrument [6].

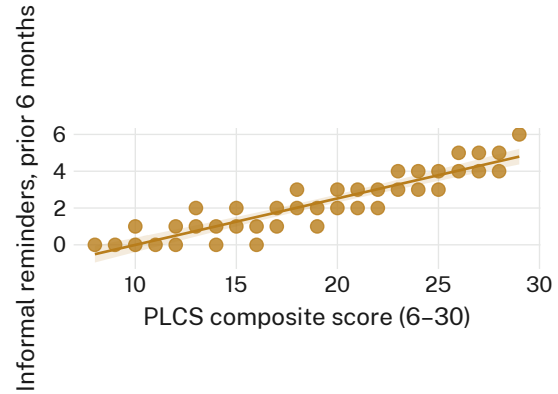


Figure 1: Composite PLCS score against each member's count of informal reminders logged over the preceding six months ($r = .52$, $p < .001$, $n = 247$).

Figure 1 plots composite PLCS score against each member's count of logged informal reminders. The two measures correlate moderately, with a member scoring at the scale's upper quartile logging, on average, more than twice the reminders of a member at the lower quartile.

Membership tenure shows no comparable relationship with perceived ceiling. Members with fewer than six months' membership ($n = 71$, median PLCS = 19, IQR 15–23) score within the same range as members with six to eighteen months' ($n = 98$, median = 20, IQR 16–24) and eighteen or more months' ($n = 78$, median = 20, IQR 17–23) tenure, and a Kruskal-Wallis test finds no reliable difference across the three groups ($H(2) = 1.14$, $p = .566$).

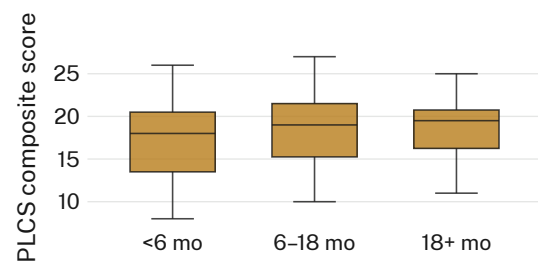


Figure 2: PLCS composite score by membership tenure. The three distributions overlap almost entirely ($H(2) = 1.14$, $p = .566$).

Figure 2 shows the three tenure groups' score distributions overlapping almost entirely.

Removing the single item most directly referencing a volunteer's manner leaves reliability essentially unchanged: alpha falls from .83 to .82, a difference well within the range expected from sampling variation alone (n.s.).

We retain the six-item form; the removed item proved the strongest single item-level predictor of the reminder count in a supplementary analysis, and dropping it would trade a negligible reliability gain for a loss of the scale’s most diagnostically specific content.

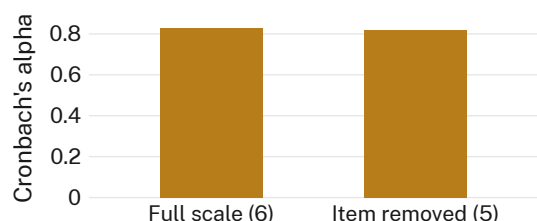


Figure 3: Cronbach’s alpha for the full six-item scale against the five-item version with the volunteer-manner item removed (.83 vs .82, n.s.).

Figure 3 reports both values.

Discussion

The PLCS’s moderate correlation with volunteers’ own reminder logs offers preliminary support for treating a member’s felt sense of an approaching limit as a proxy — imperfect but informative — for enforcement action an organisation itself is not tracking systematically. That the scale shows no association with tenure is, we suggest, the more useful finding for the organisations we studied: a ceiling this study measured appears to form quickly, within a member’s first several visits, rather than gradually across a membership. An organisation relying on an assumption that new members simply have not yet “learned how things work” would, on this evidence, be assuming a developmental process the data do not show. We observe broadly consistent factor structure and reliability across nine libraries that differ substantially in size, catalogue, and staffing model, which we take as encouraging for the instrument’s use beyond this specific sample, though the libraries were self-selected rather than randomly recruited from a wider population, and our results should be read with that qualification attached throughout.

Limitations

Our sample draws on nine toy libraries operating within a single regional network, and the reminder-log criterion depends on volunteers’ own retrospective recall of private conversations rather than a contemporaneous record —

both constraints a larger, multi-network deployment with structured logging would address. We consider the consistency of factor structure and reliability across nine libraries that differ substantially in size and catalogue an encouraging sign that the instrument is not simply fitted to one setting’s idiosyncrasies, even as we withhold any claim to representativeness beyond the network sampled. The six-month reminder window may also undercount enforcement activity for members whose tenure exceeds that window, a possibility the flat tenure effect we report makes, on balance, less rather than more likely.

Conclusion

We have introduced the Perceived Lending Ceiling Scale, a brief instrument for a phenomenon that advertised-unlimited lending schemes generate but rarely measure directly: a member’s own sense of when an unrestricted policy stops behaving like one. Initial evidence supports a single-factor structure, acceptable reliability across nine independently-run toy libraries, and moderate convergent validity against volunteers’ own informal enforcement records. Future work might deploy the PLCS, with wording adapted to the resource in question, across the wider family of nominally unlimited communal-lending schemes this network’s own prior studies have examined — street-corner book exchanges and little free libraries among them — to test whether a felt ceiling is a general feature of informally governed abundance, or a property specific to the toys themselves.

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